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REVIEW



Sindbis virus as a human pathogen—epidemiology, clinical picture and pathogenesis

Samuel Adouchief[1]*, Teemu Smura[1], Jussi Sane[2], Olli Vapalahti[1,3,4] and Satu Kurkela[1,3]

[1]*Department of Virology, Faculty of Medicine, University of Helsinki, Helsinki, Finland*
[2]*National institute for Health and Welfare (THL), Helsinki, Finland*
[3]*Department of Virology and Immunology, HUSLAB, Helsinki University Central Hospital, Helsinki, Finland*
[4]*Department of Veterinary Biosciences, Faculty of Veterinary Medicine, University of Helsinki, Helsinki, Finland*

SUMMARY

Sindbis virus (SINV; family *Togaviridae*, genus *Alphavirus*) is an enveloped RNA virus widely distributed in Eurasia, Africa, Oceania and Australia. SINV is transmitted among its natural bird hosts via mosquitoes. Human disease caused by SINV infection has been reported mainly in South Africa and in Northern Europe. Vector mosquito abundance affects the annual incidence of SINV infections with occasional outbreaks of up to 1500 patients. Symptoms include fever, malaise, rash and musculoskeletal pain. In a significant portion of patients the debilitating musculoskeletal symptoms persist for years. Chronic disease after SINV infection shares many features with autoimmune diseases. Currently there is no specific treatment available. Recently SINV infections have been detected outside the previously known distribution range. In this article we will summarize the current knowledge on epidemiology, clinical disease and pathogenesis of SINV infection in man. Copyright © 2016 John Wiley & Sons, Ltd.

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INTRODUCTION

Sindbis virus (SINV) is a widely distributed zoonotic alphavirus (family *Togaviridae*, genus *Alphavirus*) causing disease in humans. SINV is found in Eurasia, Africa and Australia, but clinical infections are reported mainly in northern Europe and South Africa. Based on the clinical disease, alphaviruses can be divided into arthritogenic viruses, namely SINV, Chikungunya (CHIKV), Ross River (RRV), Barmah Forest, Mayaro, and O'nyong nyong viruses, and into encephalitic viruses such as Western, Eastern and Venezuelan equine encephalitis viruses [1]. The SINV-induced disease is called Pogosta disease in Finland, Ockelbo disease in Sweden and Karelian fever in Russia. SINV is transmitted from its enzootic host animals to humans by mosquitoes. Symptoms include rash, fever, arthritis and myalgia. The latter two are often disabling and become prolonged in a significant portion of patients. Recent reports from Germany and northern Sweden suggest SINV to be emerging in previously unaffected regions [2–5].

In endemic areas, seroprevalence of SINV can be high, e.g. 39% in parts of Eastern Finland [6]. Prolonged musculoskeletal manifestations occur frequently [7]. In Sweden, a previous SINV infection was reported to be associated with stroke [5]. This, together with the generally low clinical recognition of the disease in both endemic and non-endemic areas, has implications for public health. Recent studies have elucidated pathology, clinical presentation and epidemiology of Sindbis virus, as well as the genetic susceptibility to clinical disease. For instance, HLA and autoantibody associations suggest that SINV induced arthritis may share features with autoimmune diseases [8].

In this review we focus on SINV as a cause of human disease, with a particular emphasis on the

*Corresponding author: S. Adouchief, Department of Virology, Faculty of Medicine, POB 21, 00014 University of Helsinki, Helsinki, Finland. E-mail: samuel.adouchief@helsinki.fi

Abbreviations used
SINV, Sindbis virus; CHIKV, Chikungunya virus; NK, Natural killer; RRV, Ross River virus; Cx., Culex; MIF, Macrophage migration inhibitory factor.

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current knowledge on epidemiology, pathogenesis and clinical manifestations of the infection. The implications of SINV phylogeny and geographical distribution will be discussed. The role of SINV as a vector in cancer therapy, as a model for viral encephalitis in mice and as an expression vector in cell biology and in vaccine development are outside the scope of this review [9–11].

VIROLOGY

Taxonomy

There are currently 31 virus species within the *Alphavirus* genus recognized in the 9th Report of the International Committee on Taxonomy of Viruses (ICTV) [12].

Phylogenetic analyses and older serological cross-reactivity studies suggest two major clusters for alphaviruses ('Old World' and 'New World' alphaviruses according to the geographical placement of the majority of the viruses within the groups), and four subgroups (Semliki Forest virus, Venezuelan Equine Encephalitis virus, Eastern Equine Encephalitis virus and Western Equine Encephalitis virus complex). SINV belongs to the Western Equine Encephalitis virus complex and is the only representative of this group outside of Americas. Notably, WEEV has been evolved as a product of an ancient recombination event between SINV-like (envelope proteins) and EEEV-like (non-structural proteins) viruses [13–16].

Phylogenetic analyses suggest a division of SINV strains into six genotypes (Figure 1) that lie in north – south axes and overlap with major bird migration routes. African and European virus strains cluster to form genotype 1, Australian, East Asian and Central Asian strains are designated genotypes 2–4 and the closely related Whataroa virus genotype 5. Strains from south-west Australia form the sixth genotype [18,25]. Sub-Saharan African and Northern European strains share a common ancestor within genotype 1 [17,26,27].

Virus structure and genome

SINV is a spherical (T = 4) virus of 70 nm in diameter. It has an icosahedral capsid of 240 monomers and a host cell derived lipid membrane with 80 trimeric spikes consisting of three heterodimers of E1 and E2 glycoproteins. The positive single-stranded RNA genome is 11.7 kB in size and has two open reading frames that code for mRNA which are generated in separate replicons for four non-structural (NSP1–4) and five structural (C, E3, E2, 6K and E1) proteins, respectively [28,29]. SINV enters target cells in a receptor mediated manner in clathrin coated vesicles. Translation and assembly of nucleocapsid occur free in the cytosol, while replication of RNA requires specific membranous replication complexes [30]. Virion assembly is completed at the plasma membrane where a lipid membrane and the glycoprotein complexes are acquired [31].

EPIDEMIOLOGY

Emergence of SINV from 1952 to present

In 1952, during an epidemic of a febrile disease with rash and arthritis in the Sindbis district near Cairo in Egypt, a new virus was isolated from mosquitoes. Serological tests suggested that this new virus, named Sindbis virus, was the causative agent of the epidemic [32]. SINV isolation from whole blood samples of patients was successful in Uganda in 1961 and from skin lesion biopsies in 1963 in South Africa [33,34].

Since its discovery, outbreaks of SINV infection have been reported in South Africa, but not in other African countries. The larger outbreaks of 1963 and 1974 coincide with the spread of SINV towards northern Europe, see Figure 2. In 1963–1964 antibodies were found in children and birds in Israel [35,54]. In 1965 the first signs of SINV in Europe were detected as antibodies against SINV were found in sera of healthy adults in Northern Italy and Finland and in birds in the Volga delta region [36,37,55]. Consistently in the phylogenetic analysis of SINV strains (Figure 1) the most recent ancestor of northern European isolates was calculated to 1961. No antibodies were found in the screening of 5000 human serum samples in Finland in 1958–1964 [38] or in Austria before 1963 [39]. In January–February 1974, a large outbreak occurred in South Africa, and later in August–September that year outbreaks were identified in Sweden and Finland. Serological surveys in the 1960s and 1970s found traces of SINV presence in Israel, Greece, Northern Italy, Turkey, Romania, Sicily and Yugoslavia [35,36,40,56–60].

Ever since the first patients were identified in Sweden in 1967 [41] and in Finland in 1974 [42], patients have been diagnosed annually in Northern Europe. In Finland in 1974–2002

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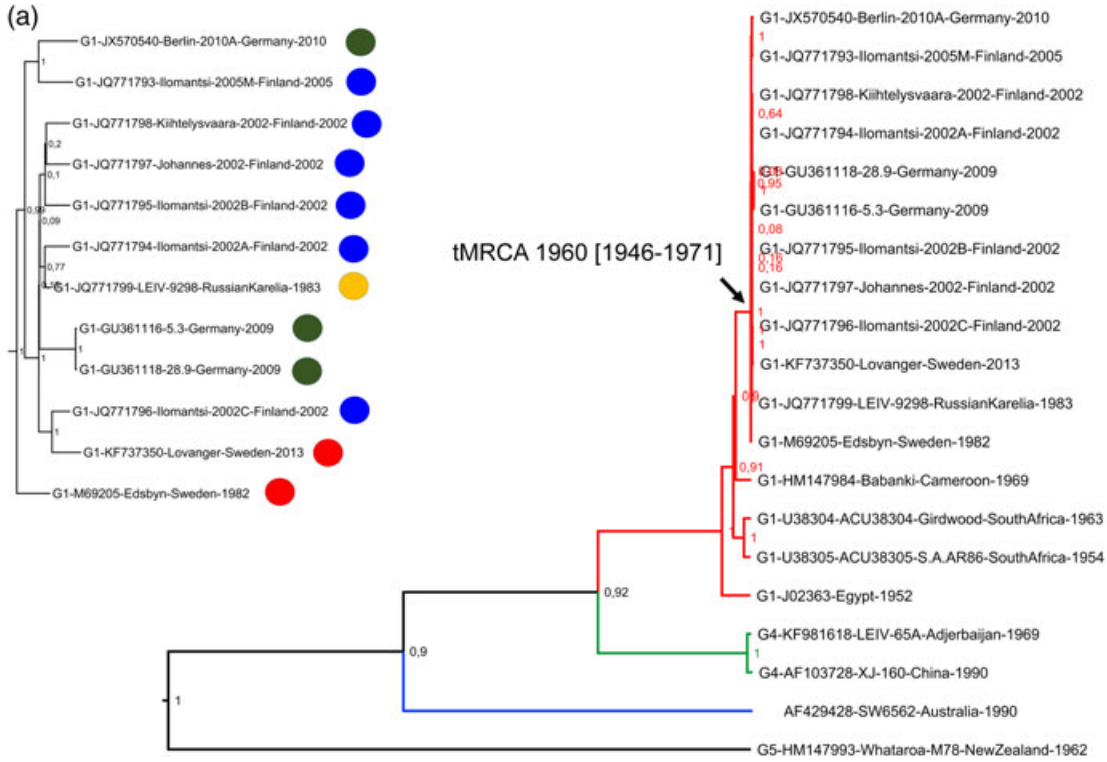


Figure 1. Maximum clade credibility tree of Sindbis virus complete coding sequence (A) and E2 sequence (B). The genotypes 1 to 5 (as defined by Lundsröm & Pfeffer 2010, [17]) and south-west Australian genotype (Saleh et al., 2003, [18]) are shown in red, orange, cyan, green, black and blue, respectively. The northern European sub-clade of genotype 1 is shown in the insert (A). The colored circles in the insert indicate the country of virus detection (Sweden, red; Finland, blue; Germany, green; Russian Karelia, orange). Posterior probabilities are shown in each node. Note, that no complete coding sequence of the genotypes 2 and 3 are currently available. All Sindbis virus sequences were downloaded from GenBank and included in the analysis. The sequences were aligned using ClustalW algorithm implemented in MEGA version 6 [19]. The best-fit substitution models (GTR + G + I for the complete coding region and K2 + G for the E2 dataset) were sought using MEGA version 6. The phylogenetic tree was constructed using Bayesian Monte Carlo Markov Chain (MCMC) method implemented in BEAST version 1.8.0 [20] using log normal relaxed molecular clock and Bayesian skyline population size demographic models [21]. The Bayesian analysis was run for 50 million states and sampled every 1000 states. Posterior probabilities were calculated with a burn-in of 5 million states and checked for convergence using Tracer version 1.6 [22]. Sindbis virus strains cluster into six distinct genotypes. As suggested earlier [17,23], within the genotype 1, the strains isolated from Sub-Saharan Africa cluster together with the strains isolated in northern Europe whereas the prototype strain from Egypt cluster together with the southern/central Europe (Slovak Republic, Italy) and Middle East (Israel, Saudi Arabia and Azerbaijan). There are, however, some exceptions to this pattern. Three strains isolated in Sweden (1985) cluster together with southern/central European and Middle-Eastern viruses with high posterior probability. On the other hand, the four strains isolated in Uganda (1960) and South Africa (1976) form a third distinct cluster within genotype 1. Within Sub-Saharan/northern European cluster, South African strains (1954–1961) form a distinct cluster whereas Babanki virus (isolated in Cameroon 1969) and northern European strains (Sweden, Finland, Norway, Russian Karelia and Germany years 1982–2013) form another cluster. On the basis of complete coding sequences the oldest sequenced northern European strain (Edsbyn, Sweden, 1982) forms an outgroup to the other northern European strains. Notably, there is no geographical pattern in the phylogeny of northern European sub-cluster. For example, the strain isolated in Berlin, Germany (2010) clusters together with a strain isolated in Ilomantsi, Eastern Finland in 2005, whereas the other nine strains (Finland, Sweden and Germany) form another highly supported cluster. Within this cluster, a strain isolated in Lovanger, north-east Sweden in 2013 [24] is related to a strain isolated from Ilomantsi, Finland in 2002. Intriguingly, on the basis of both complete coding region and E2 sequences, the estimated time of the most recent common ancestor (tMRCA) for the northern European strains dated around the year 1960, which is consistent with the timing of the first SINV-antibody detections in the Nordic countries (see the main text, chapter Epidemiology)

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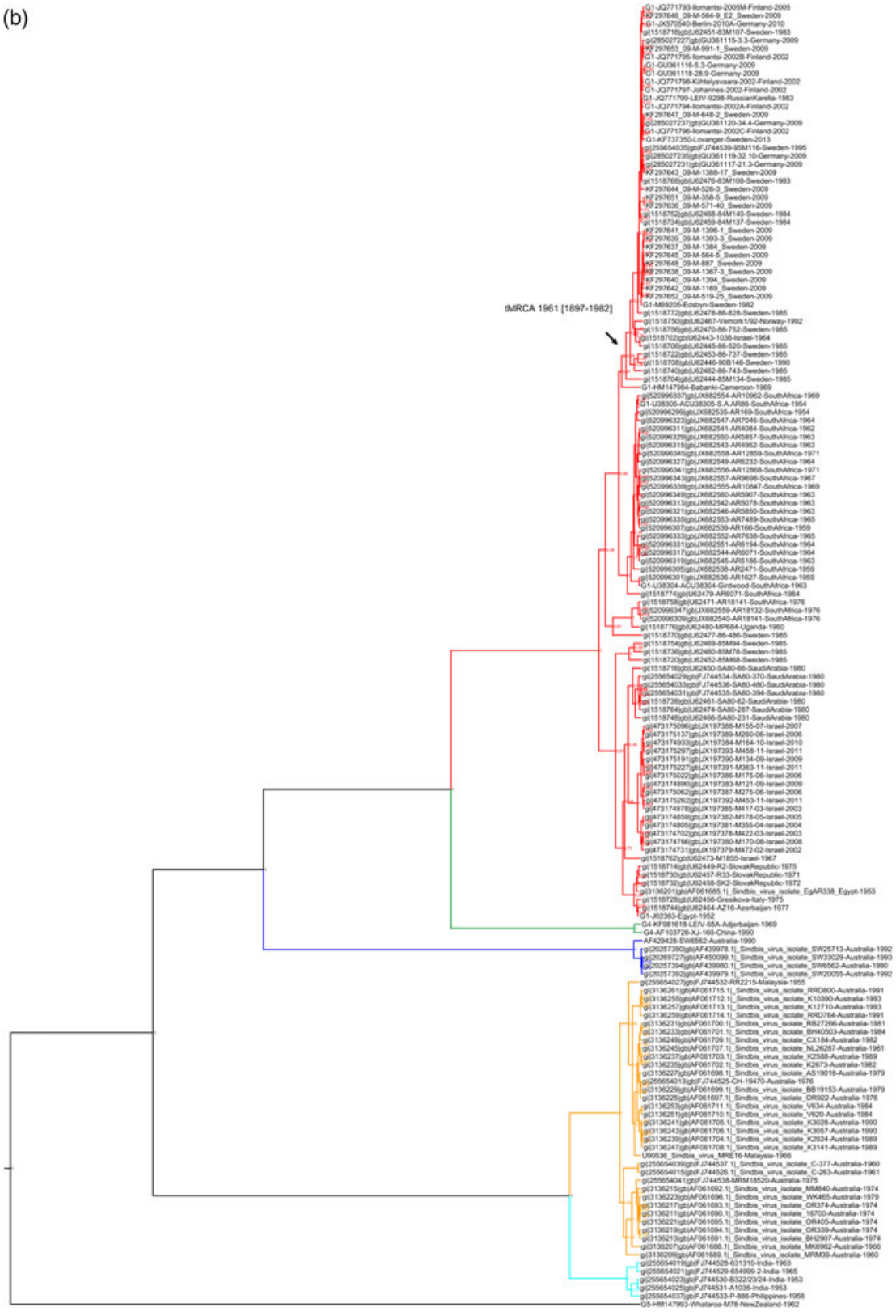


Figure1. Continued

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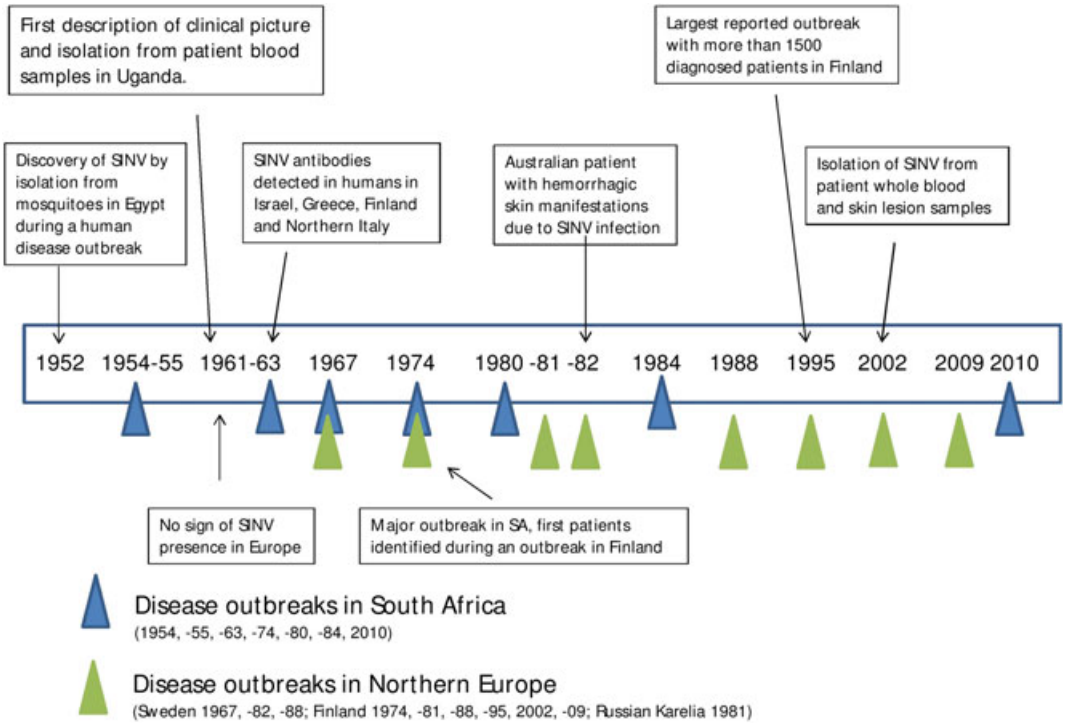


Figure 2. Timeline of selected major events in the history of Sindbis virus as a human pathogen [6,32,34–53]. Arrows indicate disease outbreaks in South Africa (blue) and in Northern Europe (green)

epidemics occurred with an interval of approximately 7 years, the largest in 1995 with 1310 diagnosed patients [6]. The expected outbreak of 2009 remained minor while in 2012 the number of diagnoses peaked to almost 200 [43]. Latest disease outbreak was observed in 2013 in the northern parts of Sweden [24].

Surveillance data implicate an increasing number of infections in Northern Europe. In Finland the average annual incidence of non-epidemic years rose from 0.4/100 000 in 1982–1987 and 0.6/100 000 in 1989–1994 to 2.1 in 1996–2001, 1.1 in 2003–2008 and to 2.0/100 000 in 2010–2012 [6,61,62]. In Sweden, the seropositivity in the northern parts of the country rose from 0.1% in 1982 to 2.9% in 2009 [5,44,63].

Signs of SINV have been reported in regions where SINV has not been detected previously. Virus isolates were obtained from mosquitoes in urban dwellings in Southern Germany (2009) [2] and in a crow in Berlin (2010) [4]. Antibodies were found in humans in Czech Republic (2002) [64] and Germany (2010) [3] and in birds in the United Kingdom (2001–2002 and 2004) [65,66] and Czech Republic (2006) [67]. Antibodies to SINV and a disease outbreak in 2013 were also detected in previously unaffected Northern Sweden, suggesting northward dispersion of SINV [5,24].

Disease incidence, antibody prevalence and seasonal distribution

As of 2014, human disease caused by SINV infection has been reported from Northern Europe, Africa, China and Australia [6,32,33,44–46,68–70], see Figure 3. Recent clinical infections have been reported from Sweden, Finland and South Africa, with outbreaks in Finland in 2009 and 2012 and in South Africa in 2010 [45,62]. While the specific reasons for the peculiar restricted disease occurrence are not well established, they are probably multifactorial and include virus ecology, clinical alert and genetic differences of both human populations and virus strains as discussed later in the article. Finland is the only country with an existing

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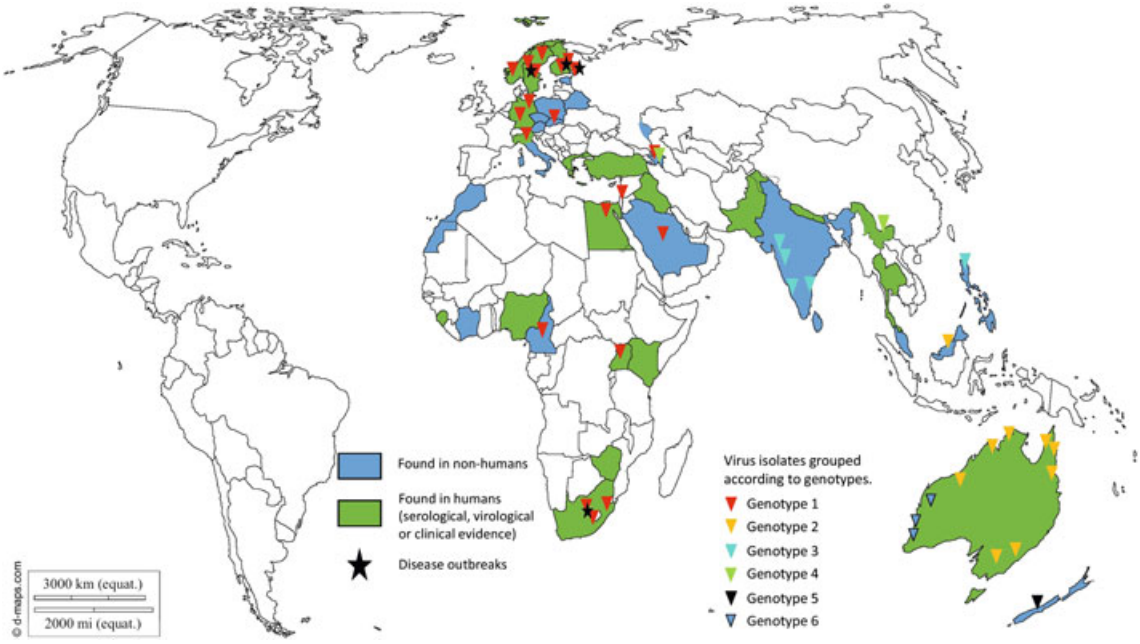


Figure 3. Geographical distribution of SINV. Regions with serological, virological or clinical evidence of human infections are marked in green. Regions with signs of SINV in non-humans (mosquitoes, birds, etc) are marked in blue. Black stars indicate regions with known disease outbreaks because of SINV infection. Colored triangles indicate virus isolates belonging to genotypes 1 to 6 shown in red, orange, cyan, green, black and blue, respectively. Base map from: http://d-maps.com

surveillance system for SINV and has the highest number of diagnosed cases worldwide with an incidence of up to 26/100 000/year in epidemic years and 2.4/100 000/year on average in non-epidemic years [43]. In South Africa, cases were reported regularly until 1980s and again since 2006, with an observation of an outbreak in 2010. A shift in the focus of public health surveillance to HIV/AIDS may explain the lack of reports between these time points [45].

In all endemic regions human infections primarily coincide with the time of most intensive primary enzootic vector mosquito activity. In Sweden and Finland this translates to a period beginning in late June and ending in early October, usually peaking in the middle of August [44,71]. In South Africa most patient cases are reported in the late summer of the southern hemisphere, March–April, around the peak of *Culex unvittatus* abundance [72].

Demographic characteristics of patients

Antibodies against SINV are more often found in men. The age-standardized seroprevalence of SINV in Finland in 1999–2003 was 6.0% for men, and 4.1% for women [61]. Investigation of 1729 individuals in Northern Sweden in 2013 showed a seropositivity rate of 3.7% in men and 2.0% in women [5].

Yet, in Northern Europe, the disease incidence appears to be higher among women. In 1995–2003, the incidence rates for women and men were 8.7 and 6.6/100 000 respectively [61]. Analysis of all serologically confirmed cases in Finland between 1984 and 2009 ($N = 3381$) showed the majority of patients to be female (58%) [71]. However in South Africa, in 2006–2010, the majority of patients were men: there were not only more samples from men (64% of all samples) but also IgM detection rate for men was higher (7% vs 5%)[45], and therefore it is difficult to estimate the disease incidence in men and women in this geographical area.

Most diagnosed patients are middle-aged adults, while adolescents are less often affected. In Finland, most diagnosed patients in 1995–2009 were those aged 45–54 years (34%) [71]. The seroprevalence in the whole population 1999–2003 was 5.2%. The

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seroprevalence increased with age reaching 15.4% in the 60–69 years age group [61]. In 2009, the incidence peaked in the 50–59 years age group (4.6/100 000/year) [73]. In Sweden, of all confirmed clinical SINV infections in 1981–1989 ($N = 245$) more than two thirds were diagnosed in those aged 30–59 years, while the proportion of adolescents (<20 years) was 6% [44]. Similarly, In South Africa in 2006–2010, the mean age of patients with SINV infection was 42 years while only 7% of patients were those aged <18 years [45].

Zoonotic cycle

Overview

Mosquitoes are the primary vectors for SINV, while birds are the main amplifying hosts [74,75]. Most virus isolates are from mosquitoes or birds but strains have been obtained also from humans, ticks, a marsh frog and a European hamster [76–80]. Factors affecting transmission of SINV and occurrence of human disease are presented in Figure 4.

Vector mosquitoes

In endemic northern European countries SINV has been isolated from *Culex torrentium/pipiens* (*Cx. torrentium/Cx. pipiens*; the hematophagic females of these two species cannot be distinguished morphologically), *Culiseta morsistans* and *Aedes cinereus* in Sweden, from a pool of mosquitoes showing highest sequence homology to *Ochlerotatus annulipes* in Finland and from *Aedes spp.* in Karelian Republic of western Russia and Norway [23,26,68,87,90]. The Aedes mosquitoes from which the Karelian strain was isolated in 1982 are since 2000 classified to *Ochlerotatus* genus [91]. In South Africa and Australia, most isolates are from *Cx. unvittatus* mosquitoes although isolates have been obtained also from *Cx. pipiens and Cx. theileri* [27]. *Cx. pipiens*, an established

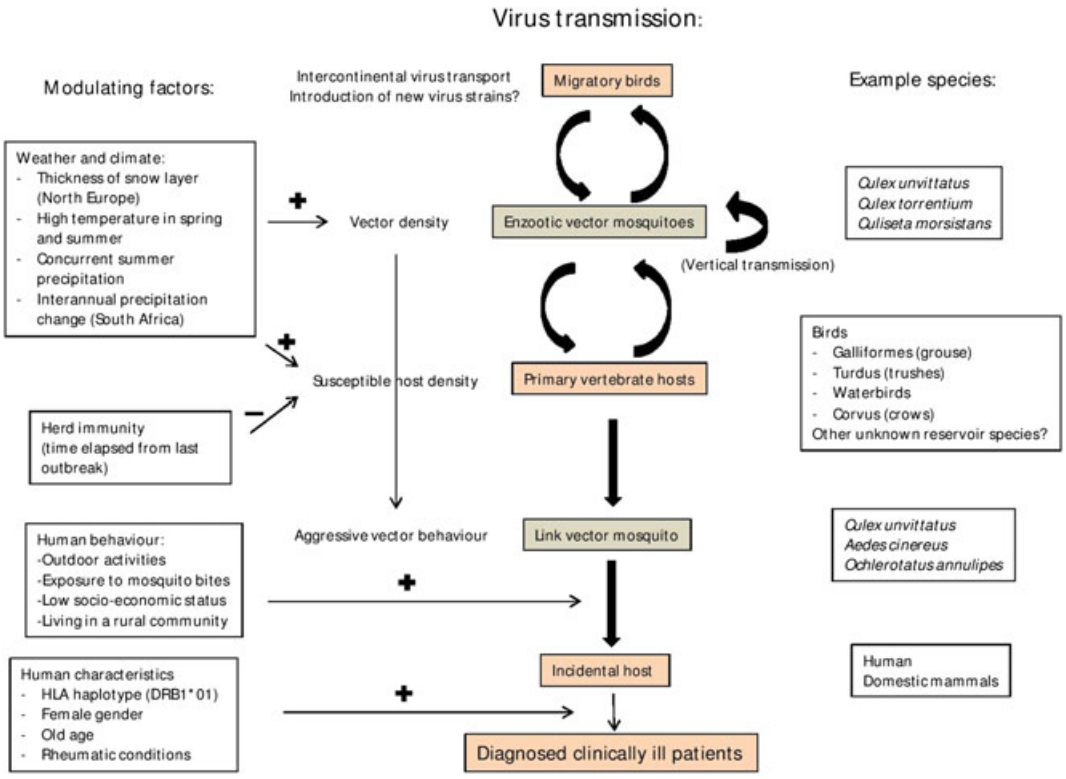


Figure 4. Factors affecting SINV transmission. In an enzootic cycle SINV is spread among its primary hosts, birds of different genera [61,66,81,82], by mosquitoes (mainly *Culex torrentium* and *Culiseta morsistans* in Northern Europe and in *Culex unvittatus* in South Africa [72,83–86]). Spill over and transmission to incidental dead end host such as human occur by both primary vector mosquitoes and by more antropophilic mosquitoes like *Aedes cinereus* or *Ochlerotatus annulipes* [86–88]. Factors influencing different steps of virus transmission are presented at left, and their effect on increasing or decreasing the probability of human disease is marked with + or − [5,6,8,61,71,89]

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vector for West Nile virus [92], has two biotopes (*Cx. pipiens* f. pipiens and *Cx. pipiens* f. molestus) with different feeding and reproduction patterns [93]; the hybrid forms might have a role in spreading viruses to humans [94].

Deducing from monthly minimum field infection rates studied in Sweden, the main enzootic vector in late spring and early summer is *Cx. torrentium* or *Cx. pipiens* and in late summer *C. morsistans* [87,95]. But as *Cx. torrentium* in comparison to *Cx. pipiens* is more common in Northern Europe [83], less affected by lower temperatures [84], more competent in transmitting SINV from bird to bird [96] and its field infection rates are higher [85] it seems evident that *Cx. torrentium* is responsible for enriching the virus among bird hosts in Northern Europe. The role of *Cx. unvittatus* as the main vector in South Africa and Australia seems equally well established as timely and geographical distribution of patient cases within South Africa and experimental virus transmission studies all support the importance of *Cx. unvittatus* rather than, for example *Cx. neavei*. Indeed both *Cx. torrentium* and *Cx. unvittatus* were shown to be excellent in transmitting SINV from bird to bird, while *Cx. pipiens* and *Cx. neavei* were considered capable but less efficient [86,96,97].

A. cinereus and *O. annulipes* are considered as most potential bridge vectors to man [88]. The possibility of *Cx. torrentium* mosquitoes being able to also transmit SINV to man should be considered.

Amplifying bird hosts

In an experimental infection study, birds of all three orders studied (passerines, galliformes and anseriformes) became highly viremic, and it was estimated that they are all capable of transmitting SINV to bridging vectors [98]. In Sweden serological studies associate passerines and especially thrushes with SINV. In 1994 the seroprevalence of passerines was 27% and merely 6% for galliformes and 4% for anseriformes [81]. In 2001, antibodies were detected in 43% of fieldfares (*Turdus pilaris*), 37% of redwings (*Turdus iliacus*) and 22% of song thrushes (*Turdus philomelos*) making these tree species the suspected main hosts in Sweden [98,99].

Yet in Finland, available evidence points towards an important role of galliformes in SINV ecology. An association between the population size of game birds and the peculiar occurrence of human disease outbreaks in seven year cycles was detected in Finland in the 1980s [6]. Furthermore, a high (65%) seroprevalence was shown among black grouse (*Tetrao tetrix*) and capercaillie (*Tetrao urogallus*) during the outbreak of 1981 and 31% one year after a human epidemic in 2002 [6,61]. In 2009 a human disease outbreak predicted according to the previous seven year interval, did not occur in the expected magnitude possibly because of an all-time smallest grouse population [62]. It has also been postulated that the number of hatch year black grouse is associated with human disease occurrence [71].

SINV antibodies have also been detected in many species of family *Corvidae*: in carrion crows (*Corvus corone*) and magpies (*Pica pica*) in England (2002) [100], and in a hooded crow (*Corvus cornix*) in Egypt (1952) [32] and Germany (2010) [4].

Migratory birds as virus carriers

Phylogeographic analysis of virus strains from different parts of the world has suggested that migratory birds might be involved in the spread of SINV to non-endemic areas [17,25]. Strains seem to be more closely related in south-north axis corresponding to bird migratory routes than in east-west axis.

Attempts to prove this hypothesis—of migratory birds disseminating SINV—by field antibody and viremia analyses from migratory birds coming to hatch in the north have brought only limited insight. In spring of 1983 in Sweden no SINV antibodies were found in birds arriving from south, yet in July of the same year 3.4% of hatching birds and 10% of birds bled in August had SINV antibodies [87]. Similarly in Finland only three sera of 806 studied (0.4%) had SINV antibodies during spring migrations of 2004 and 2005 [61]. This suggests a role for resident birds or for overwintered mosquitoes in the natural cycle of SINV in Northern Europe.

Predisposing factors for SINV infection and disease

SINV is geographically widely spread, but clinical infections are mainly reported from northern Europe and South Africa. Reasons behind the focality of the clinical disease are likely multifactorial, potentially relating to ecological circumstances,

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differences in circulating virus strains and host genetic factors.

SINV strains forming the North European – South African subcluster of genotype 1 have been linked with human disease [17]. Yet strains belonging to this subcluster have been found to circulate in wildlife also in Germany where no patients have been identified [2,4]. This could however also be explained by low clinical alert or by low human exposure (as evidenced by a low seroprevalence) to which ecological circumstances may contribute.

In Finland, the density of hatch-year black grouse [71], the thickness of snow layer in April and the precipitation and temperature in early summer associate with the occurrence and incidence of human disease [71]. In warm autumns the incidence peak shifts towards September and grows in magnitude [6]. In South Africa, high temperatures and precipitation on the previous and the current year increase the number of human cases [47,89]. Extreme weather conditions (droughts and/or floodings) often precede human outbreaks in South Africa as was the case, e.g. in 2010 [45,89]. Similarly, following the floods of 2002 first seropositive individuals were found in Czech Republic [64]. It is suggested that abundance of *Cx. torrentium* mosquitoes in Northern Europe in comparison to Central Europe could partly explain the unequal occurrence of SINV [85].

In a population-based case–control study mosquito bites were identified as the most important risk factor. In a multivariate analysis, spending time outdoors was also linked to disease risk. Mosquito repellent use had no effect probably because of lack of consistency in their use [73]. In addition living in a rural community, high percentage of population working in agriculture and high proportion of non-agricultural land, are associated with disease occurrence [71]. Living in smaller communities in the countryside and an educational level below university level were identified as factors associated with seropositivity [5].

Host genetic factors have also been associated with symptomatic SINV infection. HLA allele DRB1*01 was found to be significantly more common among SINV infected patients (23/70 alleles, 32.9%) than in the reference population (23/180 alleles, 12.8%). DRB1*01 was found particularly often in patients still suffering from persisting joint manifestations 3 years after infection (in chronic patients 9/18 (50%) alleles, vs. in totally recovered patients 14/52 (26.9%) alleles, $P = .07$) [8]. In a case study of a patient with significant persisting muscle aches after SINV infection, homozygous HLA-B*35 allele and heterozygous HLA-DRB1*01 and DRB1*04 alleles were detected [101]. In a study of 29 patients HLA B*27 and DRB1*04 alleles showed no association with the clinical disease [102].

Similar to other arthritogenic alphaviruses CHIKV and RRV, higher age and female gender have been identified as risk factors for severe and prolonged symptoms in SINV infection. However, in comparison to CHIKV, high viremia did not predict symptom prolongation [103,104].

CLINICAL FEATURES

Course of infection

The ratio of subclinical and thus undiagnosed SINV infections to all patients was estimated to be 20 or 40 to 1 in Sweden and 17 to 1 in Finland [6,44].

Median time from mosquito bite to first symptoms is four days (range 2–18 days) [73,103]. Patients seek medical care in 2 days (median) after symptom onset. During an outbreak in Finland in 2002, 6% (20/337) of patients were hospitalized. Median time spent in the ward was 4 days (range 0–47 days) [73]. Similarly, 2 of 65 patients (3%) diagnosed in Sweden during the outbreak of 1981 were hospitalized [63]. The acute phase symptoms resolve usually in two weeks, with alleviation of general symptoms and rash. The median time off work is 3 days [62].

Viremia in SINV infected humans is short and low in contrast to other arthritogenic alphaviruses [48,103,105]. The level of SINV RNA in serum of acutely ill patients was estimated to be less than 103 RNA copies/ml whereas in Chikungunya the levels may reach 10E9–10E10 RNA copies/mL [106,107]. Virus isolation from blood has been successful on two occasions [33,48].

Specific IgM antibodies become detectable within first week after infection and can be detected in practically all patients by two weeks post infection (p.i.) [103,108]. IgG becomes detectable by day 11 p.i. and its level rises for at least 6 months [103]. Peak levels of IgG are typically reached within one year p.i. [108]. It is believed that IgG antibodies evoke a lifetime protection against a new symptomatic infection. High levels of IgM persist in some patients for even 4 years and were elevated

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in 36% of 50 studied patients at 6 months p.i. [103,109]. Although no association with persisting IgM and clinical picture has been established, it is notable that in several occasions in patients with more severe symptoms, such as hemorrhage, prolonged myalgia and persisting arthritis, elevated IgM levels are detected [7,101,110].

Acute symptoms

The symptoms begin with (often itching) rash, fatigue, mild fever and headache. Some patients also have malaise, nausea, lymphadenopathy and dizziness (Table 1). Musculoskeletal symptoms appear slightly after the initial symptoms [63,103].

Skin lesions occur in practically all patients (92–100%) and are found on the trunk and the limbs [73,111]. SINV has been isolated and RNA detected from these lesions [48,112]. The rash first appears as macules that soon develop to papules (approximately 3 mm in diameter, see also Figure 5), with occasional vesicle formation [49,110]. The skin lesions are usually within the same phase [111].

Joint symptoms can be considered as one of the hallmarks of SINV disease. In 5/6 studies joint manifestations were present in nearly all acutely ill patients (95–100%). One study reported a lower percentage (35%), probably because of the retrospective study setting [45].The most commonly affected joint is the ankle (83–85%) followed by fingers, wrists and knee (56–72%). The joint pain manifests mainly as tenderness during movement but also as aching in rest. The pain resides continuously in same articulations but the occurrence of symptoms is fluctuating. Usually more than three, but in 40% of patients, at least six joints are affected [73,103].

The occurrence of muscle aches ranges from 16.4% to 62% to 100%. The patients may have difficulty in identifying the location of the pain. In a retrospective questionnaire filled in by treating physicians, myalgia was reported as more common than arthralgia [45]. Similarly, in a telephone questionnaire conducted by a single researcher, all patients reported myalgia and nearly all arthralgia [63].

Severe complications are rare. There is one report of possible meningitis [102], and one of a case with recurring hemorrhagic skin lesions [110].

Persistent sequelae

Prolonged musculoskeletal symptoms are a hallmark of arthritogenic alphaviruses. The proportion of patients with debilitating joint and muscle aches diminishes with time and the majority of patients recover completely over time (Table 2).

Quality of life and association to other medical conditions

The effect of acute SINV infection on daily routines was reported as severely restrictive by 6/73 patients (8%) and in another study as considerable or major by 24/65 patients (37%) [103,111]. Approximately 40% of patients reported incapacity for work for more than 3 days and 12% for more than a week [103]. Previous trauma to joints is associated with persisting joint pain [119].

Seropositivity for SINV was an independent predictor of having had a stroke (odds ratio 4·3, 95% confidence interval 1·4–13·0, $P = 0·011$) [5].

Moreover, pre-existing rheumatic and autoimmune associated medical conditions are more frequent among chronic than recovered patients [119].

Diagnostics

Laboratory diagnosis of SINV infection, available in a limited number of laboratories, is based on serology. Viremia is short and low and therefore PCR from serum is usually not applicable for routine diagnostics [106]. The primary method is enzyme immunoassay for detection of SINV-specific antibodies [120]. Immunofluorescence assays are suitable particularly for screening purposes. According to published reactivity of Pogosta disease patient sera with various alphaviruses [121] and our own experience, cross-reactivity between SINV (WEE complex) and the SFV complex (e.g. CHIKV) is very limited and rare in serological tests. Hemagglutination inhibition test or neutralization assays can be used for confirmation of the specificity. The detection of IgM antibodies, seroconversion and a fourfold rise in antibody titers is considered diagnostic for SINV infection [105].

Other arthritogenic alphaviruses, parvovirus B19 and non-infectious rheumatic conditions are the primary diagnostic alternatives. In Australia RRV and Barmah Forest virus are clinically well-known, while SINV is the most often isolated

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Table 1. Clinical picture of acute SINV infection [33,45,49,63,73,103,111]

	Frequency of different symptoms						
	Uganda 1961 ($n = 5$) clinical examination	**South Africa 1964 ($n = 21$) clinical examination**	**Finland 1981 ($n = 73$) questionnaire**	**Sweden 1982 ($n = 50$) tele-questionnaire**	**Finland 2002 ($n = 86$) case history, questionnaire**	**Finland 2002 ($n = 337$) questionnaire**	**South Africa 2006–2010 ($n = 58$) questionnaire to clinicians**
Symptoms	N		%	%	%	%	%
General inflammation							
Fever	4/5	Yes	23	Not prominent	41	36	67
Lymphadenopathy	1/5	No			13	17	
Dizzines					16	23	
Fatigue		Yes long			57	77	35
Paresthesia	1/5	Few	8*				
Malaise	2/5	Yes	47*	10			
Headache	5/5	Frequent	47*	14	40	49	69
Nausea		Uncommon	47*		13	18	
Musculoskeletal							
Arthritis			***	94	95	96	
Arthralgia	1/5	All	93***	20**			35
Ankle					85	83	
Knee					56	64	
Hip					23		
Finger					65		
Wrist					58	72	
Elbow					27		
Shoulder					21		
Myalgia		Occasionally	16	All	47	62	78
Skin							
Rash	0/5	All	88	96	92	96	
Itching rash			55	6	53		

Continues

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Table 1. (Continued)

	Frequency of different symptoms							
	Uganda 1961 ($n = 5$) clinical examination	South Africa 1964 ($n = 21$) clinical examination	Finland 1981 ($n = 73$) questionnaire	Sweden 1982 ($n = 50$) tele-questionnaire	Finland 2002 ($n = 86$) case history, questionnaire	Finland 2002 ($n = 337$) questionnaire	South Africa 2006–2010 ($n = 58$) questionnaire to clinicians
Eyes							
Photophobia		Not common				13	
Visual disturbances						8	

*Headache, malaise or nausea on 34/73 (47%) patients.
**Arthralgia at 3 weeks p.i.
***Reported as some joint symptoms on 68/73 (93%) patients.

alphavirus from mosquitoes [122]. In clinical practice, the diagnoses are sometimes based on clinical grounds without laboratory confirmation, especially in the endemic regions during disease outbreaks. This is however not recommended, because of a considerable risk of misdiagnosis [103].

Treatment and prevention

No specific treatment is available and currently efforts are concentrated mainly in pain alleviation with non-steroidal anti-inflammatory drugs. No vaccine is available nor in preparation at the moment.

Mosquito bites are the single most important risk factor for acquiring SINV infection and thus avoiding them with protective clothing is the preventive method of choice. Peculiarly, the use of mosquito repellents was not associated with diminished risk in Finland. It was speculated that this might be because of unaccustomed usage, as mosquitoes in general in Finland carry only few human pathogens [71,73].

PATHOGENESIS

SINV has been widely used in mouse models for encephalitic alphaviruses. However, relatively little is known about the pathogenesis of SINV-induced arthalgia/arthritis. Therefore, we also discuss here the pathogenesis of the related CHIKV and RRV, of which more conclusive information is available.

The disease manifestations caused by arthritogenic alphaviruses are thought to result from a combination of direct cellular damage caused by virus replication and indirect immune-mediated damage in the target tissues. After inoculation via mosquito bite, alphaviruses spread through the circulation to liver, spleen, muscle, lymph nodes and connective tissues around bones and joints. The viral replication is followed by an inflammatory response characterized by infiltration of lymphocytes, Natural Killer (NK) cells, neutrophils and, especially, macrophages to the target tissues. During this process, several proinflammatory cytokines and chemokines are upregulated [1,123,124].

Molecular pathologenesis

SINV enters cells by receptor-mediated endocytosis and low pH-dependent membrane fusion [125].

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Figure 5. Photographs of typical maculopapular exanthema of a patient with acute SINV infection. Photographs by courtesy of Dr. Raimo Suhonen

However, more recently a novel concept of direct cell penetration of viral genome through pores formed by viral/host proteins at plasma membrane has also been introduced [126]. The high-affinity laminin receptor, cell surface heparan sulfate and lectins DC-SIGN and L-SIGN have been reported to act as attachment factors [127–129], while an iron transporter, the natural resistance-associated macrophage protein (NRAMP2) serves as the cell surface receptor [130]. In vitro infection of mammalian cells usually leads to apoptosis [131,132], but human macrophages can also be persistently infected, which leads to their activation and increase in secretion of macrophage migration inhibitor factor (MIF), TNF-α, IL-1β, and IL-6. In the second phase, the infected macrophages start to produce matrix metalloproteinases 1 and 3 [133]. The overall cytokine response to SINV infection is still, however, weakly described.

Tissues and cell types affected by SINV infection

SINV has been isolated from human skin lesion biopsies on two occasions, and viral RNA has been detected once [34,48,112]. Examination of skin lesion biopsies showed the upper dermis as edematous with inactive T-lymphocytes and lymphoblasts near hair follicles, dermal capillaries and venules. Some macrophages but no B-lymphocytes were detectable [134]. SINV has also been shown to replicate in the skin of neonatal mice [135,136].

A muscle biopsy of a chronically myalgic patient with SINV infection revealed signs of tissue regeneration following necrosis [101]. Primary human myoblasts and myotubes are susceptible to SINV infection in vitro, yielding infective virus and showing cytopathic effect [101]. These results are consistent with those of other arthritogenic alphaviruses, because severe myositis, virus presence and myofibril pathology have been observed in human CHIKV infection [137]. SINV was experimentally shown to infect bone-associated connective tissues (periosteum, endosteum and tendons) in a mouse model [138].

Innate immunity

Alphavirus infection induces robust innate and adaptive immune responses. Levels of type 1 interferons are elevated in mouse models of SINV, CHIKV and RRV infections [139–143], and in patients with CHIKV infection [144,145]. After infection with SINV mice lacking the interferon alpha/beta and interferon gamma receptors developed a hemorrhagic fever [146]. However, alphaviruses use several mechanisms to evade interferon response.

In CHIKV- and RRV diseases the significant role of macrophages and macrophage induced cytokine response has been highlighted in patient, macaque and mouse model studies [145,147–150]. Symptoms of RRV- and CHIKV diseases are alleviated by diminishing the number of macrophages and their activity by chemical substances [151,152]. Other cell types, such as NK cells are also suspected to play a role in alphavirus pathology as NK cells have been identified in synovial samples of patients with CHIKV infection [145]. Correspondingly, in vitro SINV infected human macrophages yielded infective virus and increased the expression of matrix metalloproteinase 1 and 3 genes approximately 20-fold. The

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Table 2. The duration of arthralgia/arthritis after clinical SINV/RRV/CHIKV infection [7,102,103,108,109,113–118]

	Persistence of joint symptoms in SINV and other artritogenic virus infections												
	2002–03 Fin SINV	1998 Australia RRV	2002–03 Fin SINV	2005–2006 La Reunion CHKV	2006 India CHKV	2005–06 La Reunion CHKV	2005–06 La Reunion CHKV	2006 India Maharasht CHKV	1994 Swe SINV	1995 Fin SINV	2002–05 Fin SINV	1982–86 Swe SINV	2005–09 La Reunion CHKV
Time p.i. (years)	0.3	0.3	1	1	1	1–1.5	1.5	2	2.5	2.5	3	3–4	3
Study setting	Postal questionnaire	Clinical examination	Postal questionnaire	Clinical examination	Clinical examination	Tele quest. rheum. sym		Clinical examination	Telephone interview	Clinical examination	Clinical examination	Postal questionnaire	Clinical examination
Number of patients	65	47	66	133	315	147	88	315	16	26	49	86	180
arthritis (%)											4		
arthralgia (%)	66	68	50	87	12	57	100	5	31	31	25	24	
Mean age of symptomatic patients (years)				58 (median)	48		60	43			52		
Sex ration of symptomatic patients (M / F)				3 / 1	1 / 3		0.9/1	1 / 3			5/7		

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production of IL-1β, IL-6, TNF-α and macrophage migration inhibitory factor (MIF), a marker of macrophage activation, were also increased in this SINV study, while the occurrence of cytopathic effect was relatively mild and late [133].

Complement could also play a role in the occurrence of symptoms in alphavirus arthralgias. In RRV infections complement activation products were detected in synovial fluid of a chronically ill patient, and in inflamed tissues in a mouse model [153]. In support of this mice deficient in C3 or C3R had significantly less severe symptoms after RRV infection. Meanwhile viral clearance or leukocyte activation was not affected [153,154]. Whether this applies to SINV is uncertain. In a single patient with severe chronic myalgia the deficiency of C4B protein did not prevent symptoms [101].

Adaptive immunity

For CHIKV studies observing patients and with knockout mice models and monoclonal antibody therapies indicate a major role for humoral response but less so in RRV infection [155–159]. In mouse model studies with CHIKV CD4 + T lymphocytes were identified in arthritis development but not in virus control [156,160].

Patients with SINV infection develop a strong antibody response with especially IgG3 levels being above average [108]. IgG3 in general is present in the early phase of an infection and associates with activation of phagocytosing cells and complement [161,162]. For CHIKV a humoral response with IgG3 protects patients from chronic disease [155] and passive transfer of patient anti-CHIKV antibodies to infected mice was therapeutic [157,158]. Major human antibody responses against SINV are targeted to the structural proteins E1 and E2 [105].

Autoimmunity

Alphavirus-induced arthralgias are associated with autoimmune diseases and their markers [163–165]. SINV infected patients examined three years after infection had elevated levels of antinuclear and anti-mitochondrial autoantibodies as well as rheumatoid factor. Symptomatic SINV infection also significantly associated with HLA DRB1*01 allele [8], which is linked to rheumatoid arthritis [166] and symptomatic human parvovirus B19 infection [167].

CONCLUDING REMARKS

SINV has long been used as a model for research in molecular virology. Regardless of the widespread geographical distribution of SINV, it is only over the recent decades we have learned more of the clinical disease in humans. One of the main restrictions for epidemiological and in vivo investigations has been the geographically limited and unpredictable temporal appearance of clinical disease. The emergence of the related CHIKV infection over the past decade has prompted a new research interest for alphavirus pathogenesis in general. While most patients with SINV infection recover quickly, many of them are left with prolonged manifestations. In areas with high seroprevalence, the estimated public health burden is considerable, and further studies on the epidemiology, diagnosis and pathogenesis of human SINV infection are clearly warranted.

CONFLICT OF INTEREST

The authors have no competing interest.

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